

Accretion Disk Precession

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Consider a circular test-particle orbit at some radius R within the accretion disk that surrounds a star of mass M_a . The donor star of mass M_d orbits the accretor at radius a . Since the test particle is bound to the accretor, its orbit must lie within the accretor's Roche lobe, which has a radius of approximately

$$\frac{R_L}{a} \simeq 0.5 \left(\frac{M_a}{M_a + M_d} \right)^{1/3} . \quad (1)$$

Here we let $R = \epsilon R_L$, so that the Keplerian angular frequency of the test particle is

$$n_{\text{test}} = \left(\frac{8}{\epsilon^3} \right)^{1/2} n_{\text{orb}} , \quad (2)$$

where $n_{\text{orb}} = [G(M_a + M_d)/a^3]^{1/2}$ is the binary orbital frequency. It is often assumed that the edge of the accretion disk is at radius $\simeq 0.7 R_L$, in which case $n_{\text{test}} \simeq 4.8 n_{\text{orb}}$.

The tidal force introduced by the donor star causes the orbit of the test particle to precess in a retrograde direction around the angular momentum vector of the binary. Let $\hat{\mathbf{h}}$ and $\hat{\mathbf{H}}$ be the unit angular momentum vectors of the test particle and the binary orbit. The vector equation of motion for the precession has a simple mathematical form:

$$\frac{d\hat{\mathbf{h}}}{dt} = \gamma (\hat{\mathbf{h}} \times \hat{\mathbf{H}}) (\hat{\mathbf{h}} \cdot \hat{\mathbf{H}}) . \quad (3)$$

The characteristic frequency in this equation is given by

$$\gamma = \frac{3b}{4} \frac{n_{\text{orb}}^2}{n_{\text{test}}} , \quad (4)$$

where $b = 1 + O(R/a)^2$ is a correction to the lowest order tidal approximation; in the present context, b could be $\simeq 1.2$ – 1.8 , depending on the donor mass (*which I don't know*). Using $n_{\text{test}} \simeq 4.8 n_{\text{orb}}$, we find $\gamma \simeq 0.16 b n_{\text{orb}}$. From eq. (3), we see that $\hat{\mathbf{h}}$ makes a fixed projection onto $\hat{\mathbf{H}}$, which means that the mutual inclination I between the orbital planes of the test particle and binary is constant, as must be the case for pure precession. If Ω is the longitude of the ascending node for the test particle orbit, where the binary orbit is the reference plane, then we may write

$$\hat{\mathbf{h}} = \sin I (\sin \Omega \hat{\mathbf{x}} - \cos \Omega \hat{\mathbf{y}}) + \cos I \hat{\mathbf{H}} , \quad (5)$$

and thus

$$\begin{aligned} \frac{d\hat{\mathbf{h}}}{dt} &= \sin I \frac{d\Omega}{dt} (\cos \Omega \hat{\mathbf{x}} + \sin \Omega \hat{\mathbf{y}}) \\ &= -\frac{d\Omega}{dt} (\hat{\mathbf{h}} \times \hat{\mathbf{H}}) \\ &= \gamma (\hat{\mathbf{h}} \times \hat{\mathbf{H}}) \cos I . \end{aligned} \quad (6)$$

Therefore, the precession rate is simply

$$\frac{d\Omega}{dt} = -\gamma \cos I , \quad (7)$$

where the minus sign indicates that the precession is retrograde. For a low mutual inclination, γ is the magnitude of the precession rate.

The precession period is roughly

$$P_{\text{prec}} \simeq \frac{6.2}{b} P_{\text{orb}} . \quad (8)$$

For the case of GW Lib, where $P_{\text{orb}} \simeq 1.3$ hr, the precession period is $\simeq 8$ hr. If $b \simeq 1.5$, this period comes down to $\simeq 5$ hr. This is still longer than the mysterious 2.1 hr modulation that is observed. However, it is at least conceivable that the observed cycle is somehow related to precession, perhaps as a subharmonic of the precession period, coupled with some other nontrivial accretion disk modulation.